EXPERIMENT 4: - INFORMATION RETRIEVAL USING NLTK

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Learning Objective

To preprocess text and build a basic information retrieval system using NLTK, TF-IDF, and cosine similarity.

Problem Statement

To retrieve relevant Amazon customer reviews based on a user's search query using text similarity.

TASKS TO PERFORM

1. Load the Amazon Fine Food Reviews dataset into a Pandas DataFrame.
2. Import the required Python libraries and NLTK resources.
3. Remove missing or null reviews and duplicate records.
4. Select the review text column and preprocess the text.
5. Apply preprocessing to all reviews and store them in a new column named Cleaned_Text.
6. Display the original and cleaned versions of five reviews.
7. Convert cleaned reviews into TF-IDF vectors and display the matrix shape and number of unique terms.
8. Create a retrieval function using cosine similarity to return the top five relevant reviews.
9. Test the retrieval system using the sample queries: "great product with fast shipping", "disappointed", "excellent taste", and "poor packaging". Compare the relevance of the results.

AIM

To develop an information retrieval system using NLTK, TF-IDF vectorization, and cosine similarity to retrieve relevant Amazon customer reviews.

ALGORITHM

1. Import the required libraries and load the dataset.
2. Remove missing reviews and duplicate records.
3. Clean text by removing punctuation, numbers, and stopwords.
4. Apply stemming to the cleaned reviews.

5. Convert reviews into TF-IDF vectors.
6. Calculate cosine similarity between the query and reviews.
7. Retrieve and display the top five relevant reviews for each query.
8. Calculate cosine similarity between the query and reviews, retrieve the top five matching reviews, and display their similarity scores and original text.
9. Test the retrieval system using the queries "great product with fast shipping", "disappointed", "excellent taste", and "poor packaging".
10. Compare the retrieved reviews for each query and comment on their relevance.

PROGRAM AND OUTPUT

```
# Question 1: Install and import the required Python libraries.
# LUQMAN T (241801141)
!pip -q install nltk scikit-learn pandas
import pandas as pd
import numpy as np
import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

```
# Question 2: Load the Amazon Fine Food Reviews dataset into a Pandas DataFrame.
# LUQMAN T (241801141)
url = "https://huggingface.co/datasets/PJ2005/amazon-fine-food-reviews/resolve/main/Reviews.csv"
df = pd.read_csv(url, nrows=10000)
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
display(df.head())
```

```
Shape: (10000, 10)
Columns: ['Id', 'ProductId', 'UserId', 'ProfileName', 'HelpfulnessNumerator', 'HelpfulnessDenominator', 'Score', 'Time', 'Su
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time	Summ
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1	1	5	1303862400	G Qu Dog F

```
# Question 3: Import the required NLTK resources.
# LUQMAN T (241801141)
nltk.download("stopwords", quiet=True)
stop_words = set(stopwords.words("english"))
stemmer = PorterStemmer()
print("NLTK stopwords loaded:", len(stop_words))
```

```
NLTK stopwords loaded: 198
```

```
# Question 4: Select the review text column, remove missing/null reviews and duplicates, and use the first 10,000 reviews.
# LUQMAN T (241801141)
df = df.dropna(subset=["Text"]).drop_duplicates().copy()
df["Text"] = df["Text"].astype(str)
df = df.head(10000).copy()
print("Rows after cleaning:", len(df))
print("Missing review texts:", df["Text"].isna().sum())
```

```
Rows after cleaning: 10000
Missing review texts: 0
```

```
# Question 5: Create a preprocessing function to lowercase text, remove punctuation/numbers/special characters, tokenize, r
# LUQMAN T (241801141)
def preprocess(text):
    text = re.sub(r"<.*?>", " ", str(text).lower())
    text = re.sub(r"[^a-z\s]", " ", text)
    words = [stemmer.stem(w) for w in text.split()
             if w not in stop_words and len(w) > 1]
    return " ".join(words)

print(preprocess("Great product with fast shipping!"))
```

```
great product fast ship
```

```
# Question 6: Preprocess all reviews and store the results in a new Cleaned_Text column.
# LUQMAN T (241801141)
df["Cleaned_Text"] = df["Text"].apply(preprocess)
print("Preprocessing completed for", len(df), "reviews.")
display(df[["Text", "Cleaned_Text"]].head())
```

```
Preprocessing completed for 10000 reviews.
```

	Text	Cleaned_Text
0	I have bought several of the Vitality canned d...	bought sever vital can dog food product found ...
1	Product arrived labeled as Jumbo Salted Peanut...	product arriv label jumbo salt peanut peanut a...
2	This is a confection that has been around a fe...	confect around centuri light pillowi citru gel...
3	If you are looking for the secret ingredient i...	look secret ingredi robitussin believ found go...
4	Great taffy at a great price. There was a wid...	great taffi great price wide assort yummi taff...

```
# Question 7: Display the original and cleaned versions of any five reviews.
# LUQMAN T (241801141)
for i, row in df[["Text", "Cleaned_Text"]].head(5).iterrows():
    print("\nOriginal:", row["Text"][:400])
    print("Cleaned:", row["Cleaned_Text"][:400])
    print("-" * 50)
```

```
Original: I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The
Cleaned: bought sever vital can dog food product found good qualiti product look like stew process meat smell better labrad
--------------------------------------------------

Original: Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if thi
Cleaned: product arriv label jumbo salt peanut peanut actual small size unsalt sure error vendor intend repres product jumb
--------------------------------------------------

Original: This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with nuts - in t
Cleaned: confect around centuri light pillowi citru gelatin nut case filbert cut tini squar liber coat powder sugar tini mo
--------------------------------------------------

Original: If you are looking for the secret ingredient in Robitussin I believe I have found it.  I got this in addition to t
Cleaned: look secret ingredi robitussin believ found got addit root beer extract order good made cherri soda flavor medicin
--------------------------------------------------

Original: Great taffy at a great price.  There was a wide assortment of yummy taffy.  Delivery was very quick.  If your a ta
Cleaned: great taffi great price wide assort yummi taffi deliveri quick taffi lover deal
--------------------------------------------------
```

```
# Question 8: Convert cleaned reviews into TF-IDF vectors and display matrix shape and number of unique terms.
# LUQMAN T (241801141)
vectorizer = TfidfVectorizer(max_features=5000)
tfidf_matrix = vectorizer.fit_transform(df["Cleaned_Text"])
print("TF-IDF matrix shape:", tfidf_matrix.shape)
print("Number of unique terms:", len(vectorizer.get_feature_names_out()))
```

```
TF-IDF matrix shape: (10000, 5000)
Number of unique terms: 5000
```

```
# Question 9: Create a retrieval function that preprocesses a query and converts it into a TF-IDF vector.
# LUQMAN T (241801141)
def prepare_query(query):
    return vectorizer.transform([preprocess(query)])

print("Query preprocessing function is ready.")
```

```
Query preprocessing function is ready.
```

```
# Question 10: Calculate cosine similarity and return the top five reviews with similarity scores and original text.
# LUQMAN T (241801141)
def retrieve_reviews(query, top_n=5):
    query_vector = prepare_query(query)
    scores = cosine_similarity(query_vector, tfidf_matrix).ravel()
    indices = scores.argsort()[-top_n:][::-1]
    result = df.iloc[indices][["Summary", "Text"]].copy()
    result["Similarity_Score"] = scores[indices]
    return result

display(retrieve_reviews("great product with fast shipping"))
```

	Summary	Text	Similarity_Score
5226	Great product and would buy again	Enjoyed the product and they also provided ver...	0.576027
8021	good tea	The tea is good and fresh. We enjoy it. The sh...	0.522053
5417	Love these meals	My son really enjoys all these flavors- eats t...	0.476972
5057	Amazingly Good!	These are very good, they were a great price a...	0.441529
6034	energy drink	The energy drink is a great product. The shipp...	0.424585

```
# Question 11: Test the retrieval system using the four sample queries.
# LUQMAN T (241801141)
queries = [
    "great product with fast shipping",
    "disappointed",
    "excellent taste",
    "poor packaging"
]
for query in queries:
    print("\nQuery:", query)
    display(retrieve_reviews(query, top_n=5))
```

```
Query: great product with fast shipping
```

	Summary	Text	Similarity_Score
5226	Great product and would buy again	Enjoyed the product and they also provided ver...	0.576027
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6034	energy drink	The energy drink is a great product. The shipp...	0.424585

```
Query: disappointed
```

	Summary	Text	Similarity_Score
3151	Bread	I am a bit disappointed. The flavor was not w...	0.524904
4295	Better mixes out there!	When I saw the ingredients on this box I knew ...	0.348591
4378	good product, too expensive	The product is very good. Way too expensive an...	0.341480
788	disappointment	I was disappointed in this product because I t...	0.340374
6548	Different	Disappointed. The big boxes had a very differ...	0.339557

```
Query: excellent taste
```

	Summary	Text	Similarity_Score
2332	Excellent	It is very very good sweets Amazon always...	0.561229
3407	Grove Square K-cups	We recieved the product on time and in excelle...	0.502545
1743	Popchips	The Popchips were excellent - had excellent fl...	0.499514
9503	Senseo coffee- French Vanilla Caramel	Excellent tasting. Will have to remember to o...	0.494205
7324	Purchased chamomile	Excellent product. Exactly what they said it w...	0.438834

```
Query: poor packaging
```

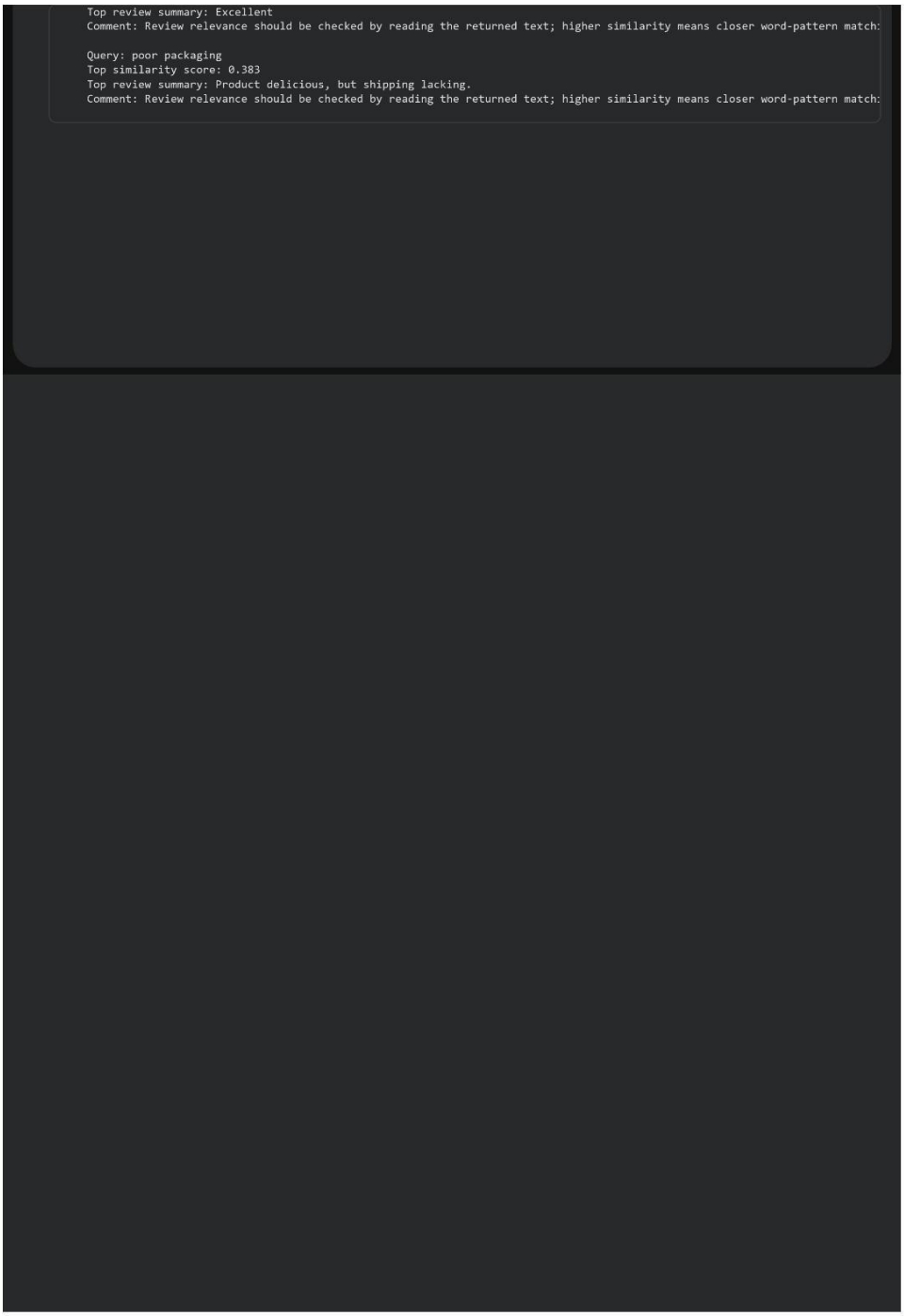
	Summary	Text	Similarity_Score
6185	Product delicious, but shipping lacking.	I was delighted to find this product online, b...	0.382790
2006	Maybe it was just my batch?	Didnt really enjoy these after wanting to try ...	0.329255
7579	wonderful soup BUT....	The soup is rich and tasty, but DON'T buy it f...	0.323083
5032	Very poor service	I would agree with the other poor ratings for ...	0.293346
532	Great price, but not as tangy as I expected.	Chips are not as tangy as I hoped or expected....	0.292871

```
# Question 12: Compare retrieved reviews for each query and comment on their relevance.
# LUQMAN T (241801141)
for query in queries:
    results = retrieve_reviews(query, top_n=5)
    print("\nQuery:", query)
    print("Top similarity score:", round(results["Similarity_Score"].max(), 3))
    print("Top review summary:", results.iloc[0]["Summary"])
    print("Comment: Review relevance should be checked by reading the returned text; higher similarity means closer word-pa
```

```
Query: great product with fast shipping
Top similarity score: 0.576
Top review summary: Great product and would buy again
Comment: Review relevance should be checked by reading the returned text; higher similarity means closer word-pattern match

Query: disappointed
Top similarity score: 0.525
Top review summary: Bread
Comment: Review relevance should be checked by reading the returned text; higher similarity means closer word-pattern match

Query: excellent taste
Top similarity score: 0.561
```



RESULT

Customer reviews were processed using NLTK to extract tokens, lemmas, POS tags, and named entities. The extracted information was displayed in a structured format.